

SUVARNABHUMI INTL, Thailand

WMO: 484290

Lat: 13.6864N

Lon: 100.7675E

Elev: 20

StdP: 14.69

Time Zone: 7.00 (E07)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 66.7	(c) 69.7	(d) 50.5	(e) 54.4	(f) 75.3	(g) 53.5	(h) 61.0	(i) 77.5	(j) 16.8	(k) 76.7	(l) 15.8	(m) 80.1	(n) 6.4	(o) 30	(p) 0.415

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 4	(b) 12.2	(c) 96.2	(d) 77.5	(e) 95.0	(f) 77.5	(g) 93.7	(h) 77.3	(i) 81.0	(j) 89.9	(k) 80.2	(l) 89.2	(m) 79.7	(n) 88.6	(o) 9.9	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.8	149.6	86.4	77.9	145.2	85.6	77.3	142.0	85.0	44.7	89.6	43.9	89.4	43.3	89.0	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 16.7	(b) 15.1	(c) 13.7	(e) 63.2	(f) 98.8	(g) 3.6	(h) 1.9	(i) 60.6	(j) 100.1	(k) 58.5	(l) 101.2	(m) 56.5	(n) 102.2	(o) 53.9	(p) 103.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	84.3	80.6	83.4	85.6	87.6	87.2	85.6	84.5	84.5	84.2	84.0	83.2	80.8
	DBStd	3.39	3.70	2.61	2.90	2.73	2.83	1.99	2.02	1.83	1.99	2.06	2.68	3.87
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12502	947	935	1102	1129	1152	1069	1069	1069	1025	1055	996	954
	CDD65	7027	482	515	637	679	687	619	604	604	575	590	546	489
	CDH74	86056	4904	5875	8096	9191	9404	8210	7521	7511	6852	6982	6354	5156
	CDH80	38502	1770	2378	3859	4988	5017	3969	3231	3217	2783	2798	2561	1931
Wind	WSAvg	6.7	5.9	6.9	8.0	7.8	7.2	7.4	7.2	7.1	6.1	5.4	5.8	5.9
Precipitation	PrecAvg	58.5	0.7	0.7	1.9	3.3	7.6	7.0	6.6	7.8	12.0	9.1	1.5	0.3
	PrecMax	76.5	3.4	2.6	6.1	8.6	15.6	12.8	10.5	13.7	18.5	16.5	5.6	1.7
	PrecMin	40.0	0.0	0.0	0.2	0.6	2.4	2.4	1.7	3.7	8.0	2.2	0.0	0.0
	PrecStd	8.7	0.8	0.7	1.7	2.4	3.3	2.6	2.4	3.0	2.5	3.1	1.5	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.4	94.0	95.3	98.1	98.1	96.1	93.6	94.4	94.0	93.3	93.4	93.1
		MCWB	73.2	72.3	74.6	77.4	78.2	77.7	77.2	76.7	77.5	77.3	75.9	75.4
	2%	DB	90.8	92.1	93.8	96.4	96.2	93.5	91.9	92.0	92.2	91.8	91.8	91.4
		MCWB	73.1	73.8	75.7	77.5	78.2	78.0	77.2	77.2	77.6	77.1	75.6	74.6
	5%	DB	89.5	90.8	92.5	95.0	94.8	91.8	90.8	90.6	90.9	90.8	90.7	89.9
		MCWB	72.7	74.5	76.2	77.8	78.2	77.9	77.1	77.2	77.4	77.0	75.4	74.0
	10%	DB	88.0	89.5	91.4	93.6	93.2	90.6	89.6	89.4	89.5	89.5	89.4	88.3
		MCWB	72.4	74.7	76.4	77.9	78.1	77.7	77.0	77.0	77.2	76.9	75.0	73.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.2	79.5	80.4	82.4	81.9	81.0	80.0	79.9	80.1	80.1	79.2	78.4
		MCDB	85.9	87.5	88.5	91.3	91.2	88.7	87.9	87.5	88.7	87.7	87.7	86.3
	2%	WB	77.3	78.3	79.2	81.2	80.8	80.1	79.1	79.1	79.1	79.2	78.3	77.4
		MCDB	84.6	85.9	88.1	90.3	90.4	88.3	87.2	87.0	87.6	87.1	86.5	85.4
	5%	WB	76.5	77.5	78.5	80.2	80.1	79.4	78.4	78.6	78.6	78.7	77.7	76.7
		MCDB	83.9	85.5	87.5	89.6	89.7	87.8	86.6	86.6	86.9	86.6	85.9	84.8
	10%	WB	75.7	77.0	78.0	79.4	79.5	78.9	78.0	78.1	78.1	78.1	77.0	75.7
		MCDB	83.2	85.2	87.0	88.9	89.1	87.5	86.3	86.2	86.3	86.0	85.3	84.1
Mean Daily Temperature Range	5% DB	MDBR	14.5	13.2	11.8	12.2	11.6	10.4	10.2	10.6	11.0	10.8	11.9	13.9
		MCDBR	15.2	14.1	12.9	12.9	12.5	12.0	11.5	11.9	12.6	12.3	12.9	14.4
		MCWBR	4.2	4.1	3.8	3.8	3.8	4.2	4.2	4.3	4.1	3.6	3.6	4.0
	5% WB	MCDBR	12.2	11.7	11.4	11.5	11.7	11.1	10.6	11.1	11.6	11.5	11.8	12.8
		MCWBR	3.9	3.6	3.7	3.8	4.2	4.2	4.2	4.4	4.1	3.9	3.9	3.9
Clear-Sky Solar Irradiance	taub	0.543	0.613	0.637	0.577	0.501	0.470	0.463	0.464	0.488	0.574	0.519	0.504	
	taud	1.960	1.819	1.795	1.978	2.246	2.331	2.352	2.334	2.259	1.974	2.108	2.097	
	Ebn at Noon	240	229	228	241	256	261	264	266	262	236	246	248	
	Edh at Noon	57	69	71	59	45	41	40	41	44	58	49	49	
All-Sky Solar Radiation	RadAvg	1607	1731	1774	1893	1726	1649	1575	1590	1541	1513	1560	1555	
	RadStd	95	75	126	115	117	83	80	98	85	85	104	74	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (32 neighbors)	N/A	N/A	N/A	N/A	+0.36	+0.38	N/A	N/A	+126	+126

Nomenclature: See separate page